



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R)

A.Y. 2026 - 2027

TITLE OF THE PROJECT		Verifiable Chain-of-Custody for Digital Evidence Files	
S. No.	University ID	Name	
1	2320030360	Chilakapati Srijaya Chakradhar	
2	2320030195	Gogineni Nikhil Sai	
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Name of the Guide		Dr. Archana Kalidindi	

Abstract

Digital forensic investigations depend on the integrity of evidence artifacts like disk images, memory dumps, browser caches, and log exports throughout their lifecycle from collection to analysis, yet conventional evidence-handling workflows offer no cryptographic guarantee of that integrity: once a file is stored or passed between analysts, there is no verifiable proof that it remains unaltered, nor an immutable record of who accessed or handled it and when. This project addresses that gap with a Verifiable Chain-of-Custody system built from first principles, in which each evidence file is hashed on intake and recorded as a leaf in a Merkle tree for efficient proof-of-inclusion verification, custody events such as transfer, access, or annotation are logged as ECDSA-signed, hash-linked entries forming an append-only chain without reliance on distributed consensus or public blockchain infrastructure, and an accompanying verifier tool independently recomputes the hash chain and Merkle proofs to confirm whether evidence or its custody history has been altered since collection, offering a lightweight, dependency-minimal approach to strengthening evidentiary trust in digital forensics.

Signature of the Guide

HOD